



CHILLER CARRIER 30HK115

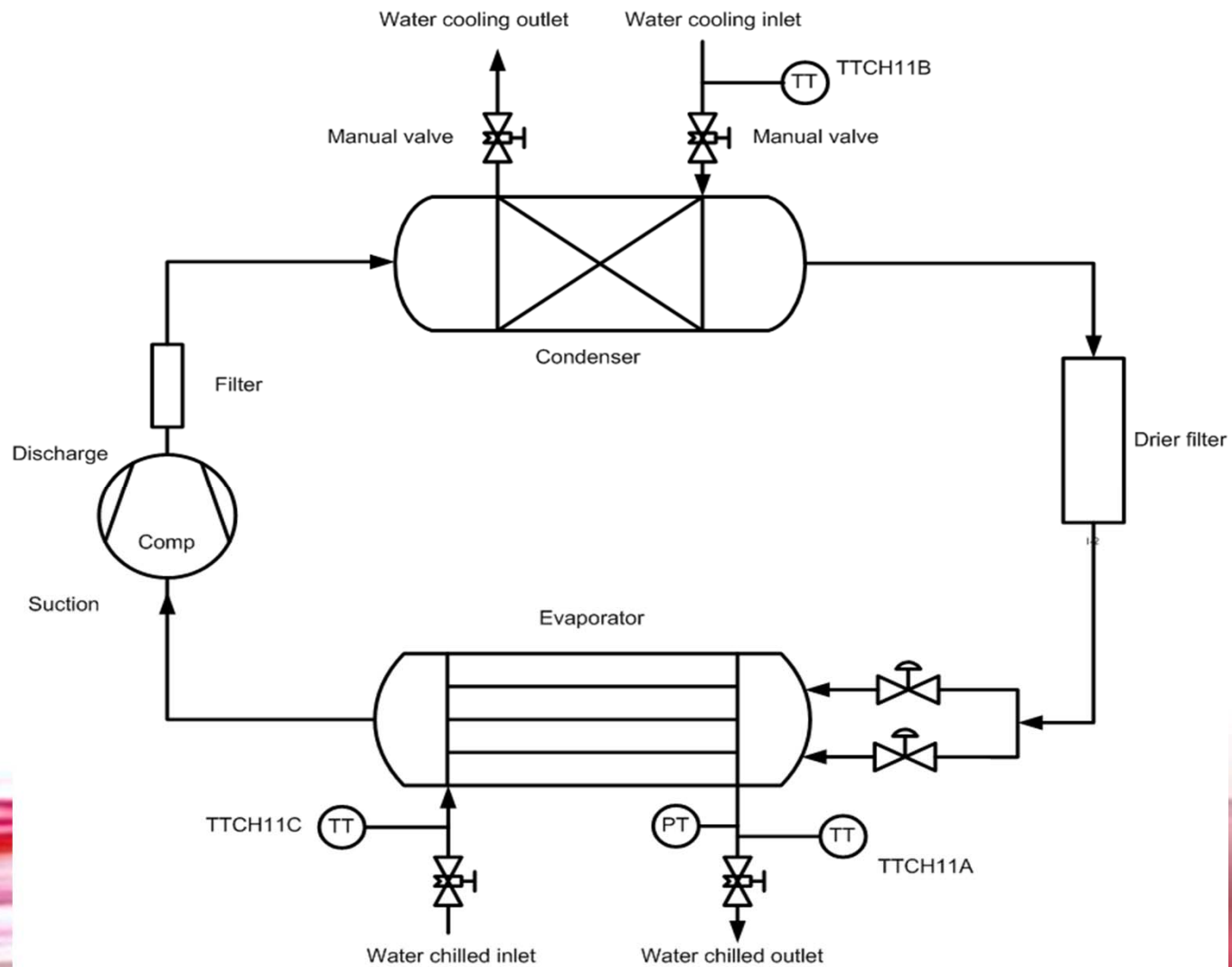
By Eduward Tigor H.L

E & I Departement

Aplikasi chiller PT WINA GRESIK

- Finishing : Water Cooled Chiller
- Fraksinasi 2500 : Absorption Chiller
- Fraksinasi 600 : Water Cooled Chiller
- Biodiesel : Air Cooled chiller
- CPC : Water Cooled Chiller
- CPKO : Absorption Chiller (future)

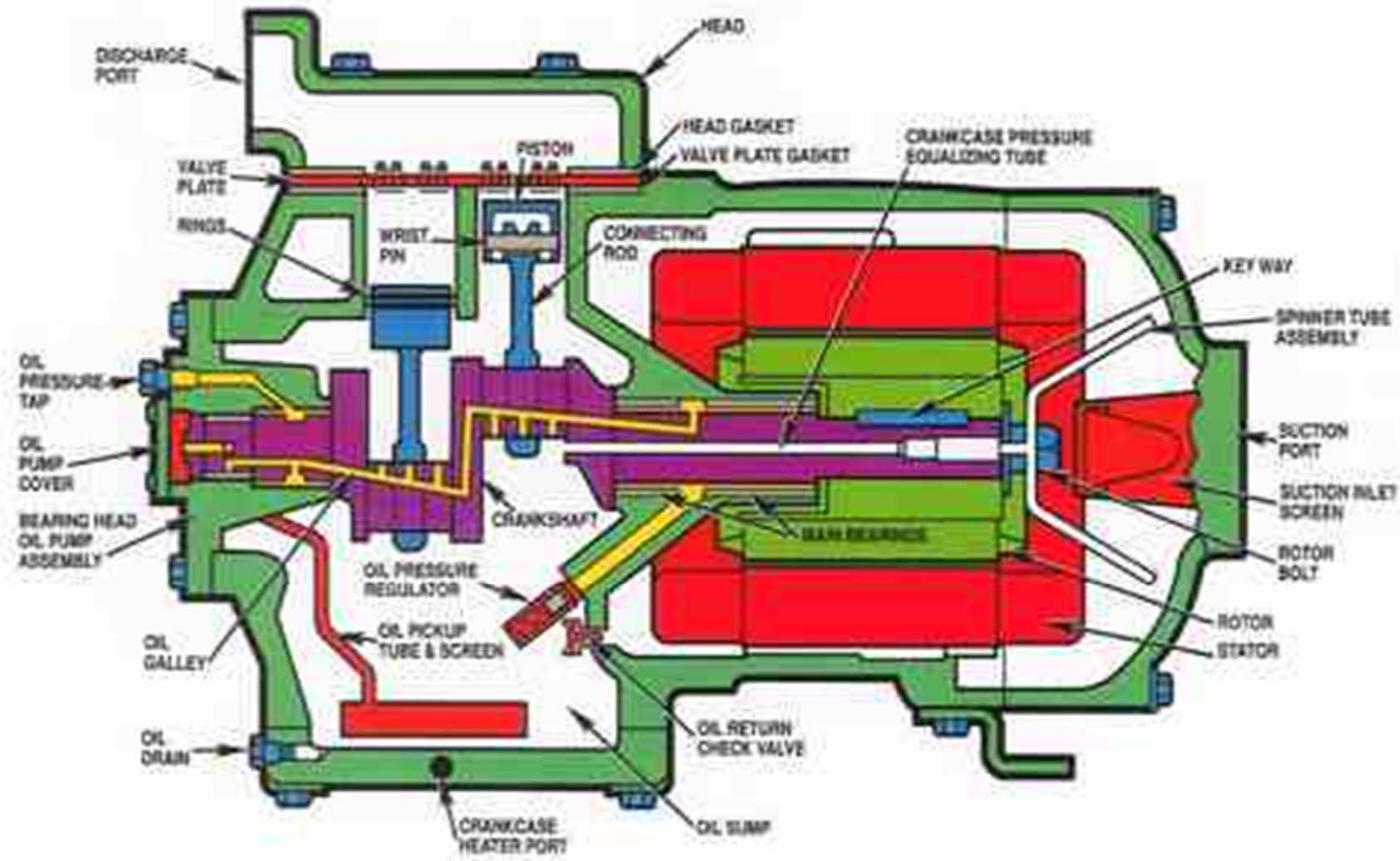
Diagram Blok



Prinsip Kerja

- Gas freon R-22 dikompresi dari evaporator ke kondenser mengakibatkan gas freon mengalami perubahan fasa dari gas ke cair, dan perubahan temperatur. Dikondenser freon diturunkan temperaturnya dengan prinsip perpindahan kalor oleh water cooling diteruskan ke evaporator kembali setelah melewati drier filter. Dievaporator pada keadaan vacuum mengalami perubahan fasa dari cair ke gas, penguapan ini mengakibatkan gas freon dapat menyerap panas dari water chilled.

Compressor



Finishing Plant Chiller

- Finishing Plant Chiller: Brand-**Carrier 30HK**

115

- Aplikasi: mesin ploder, vacuum pump, Tree roll mill
- Type : Water Cooling
- Refrigrant: R-22

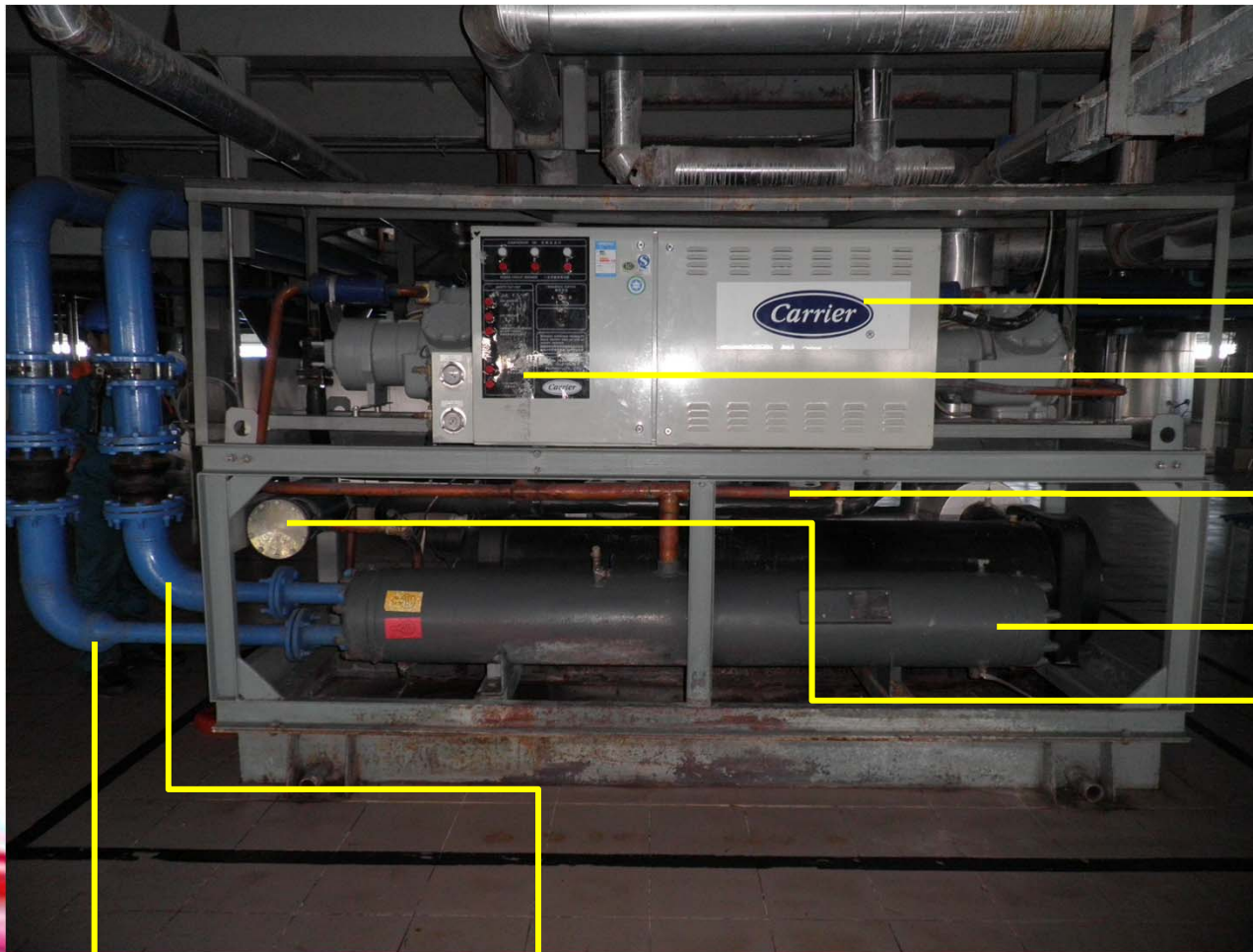
Chiller Type	Compressor Type	Refrigerant
Absorption	Single-Effect, Steam-Fired	Water
	Single-Effect, Hot Water	
	Double-Effect, Direct-Fired	
	Double-Effect, Steam-Fired	
Rotary	Hermetic Centrifugal	R-134a
	Hermetic Screw	R-134a
Packaged Air-Cooled	Scroll	R-22
	Scroll	Puron®
	Scroll	Puron®
	Screw	R-134a
Packaged Water-Cooled	Reciprocating	R-22
		R-22
		R-22
	Screw	R-134a
Packaged Condenserless	Reciprocating	R-22
		R-22
	Screw	R-134a

Spesifikasi Carrier 30 HK 115

- Nominal Cooling/Heating capacity : 344,4 KW
- Voltage/Freq/Phase : 380 V/50 Hz/ 3 Phase
- Cooling/Heating Rate Current : 3 x 68 A
- Model Num : 30 HK 115
- Rate Power Output Motor : 86,4 KW
- Chilled Water Leaving Temp : 7 °C
- Cooling Water Entering Temperature : 30 °C
- Flow of cooler/Condenser : 59.24/74.00 T/h
- Refrigerant : R 22
- Refrigerant W.T : 63 Kg
- Rpm : 1450 r/min
- W.T : 1920 kg

Carrier 30HK 115

- Tampak depan



Control Panel

Alarm Indikator

High pressure line

Condenser

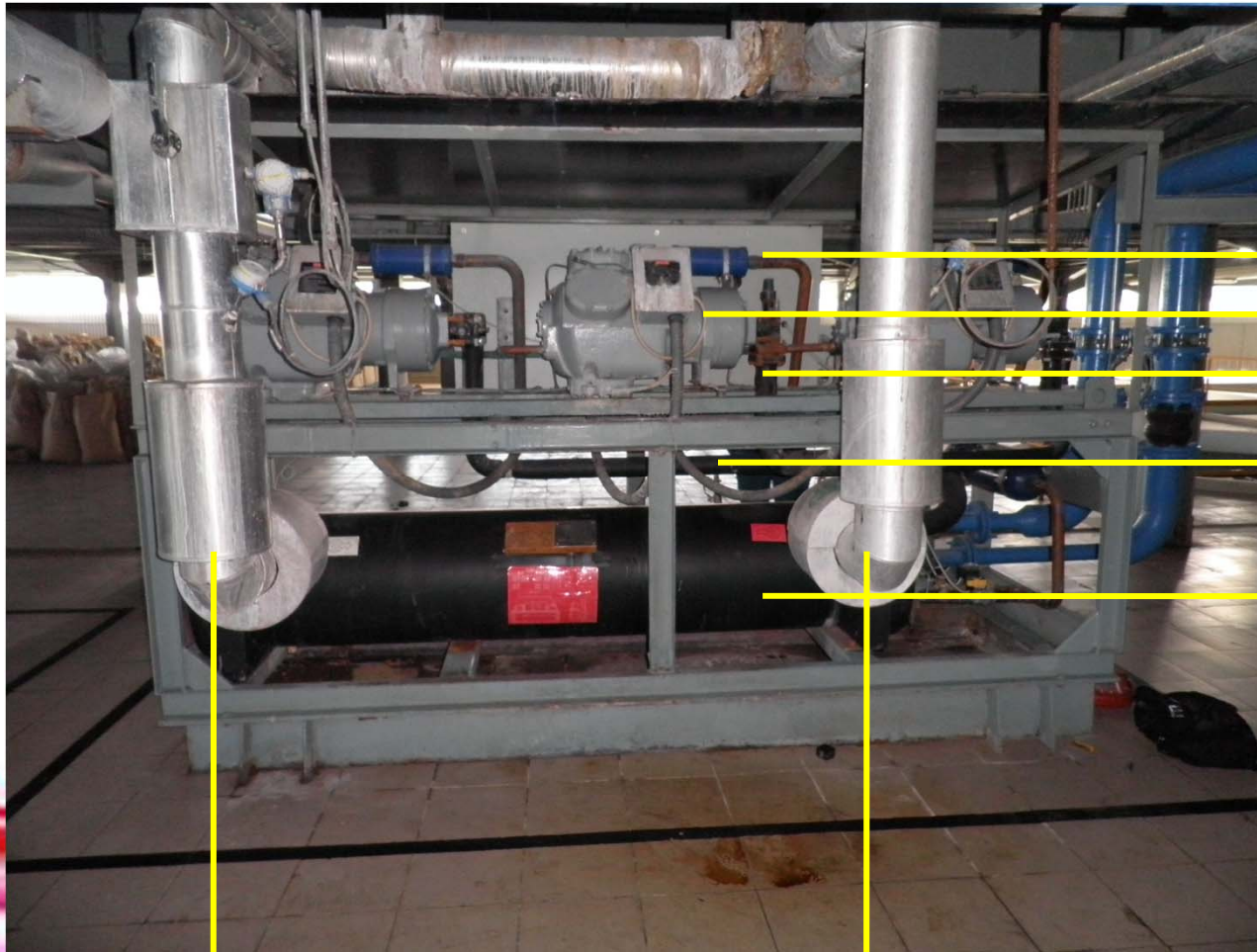
Dry Filter

Water cooling inlet

Water cooling outlet

Carrier 30HK 115

- Tampak Belakang



Discharge

Compressor

Suction

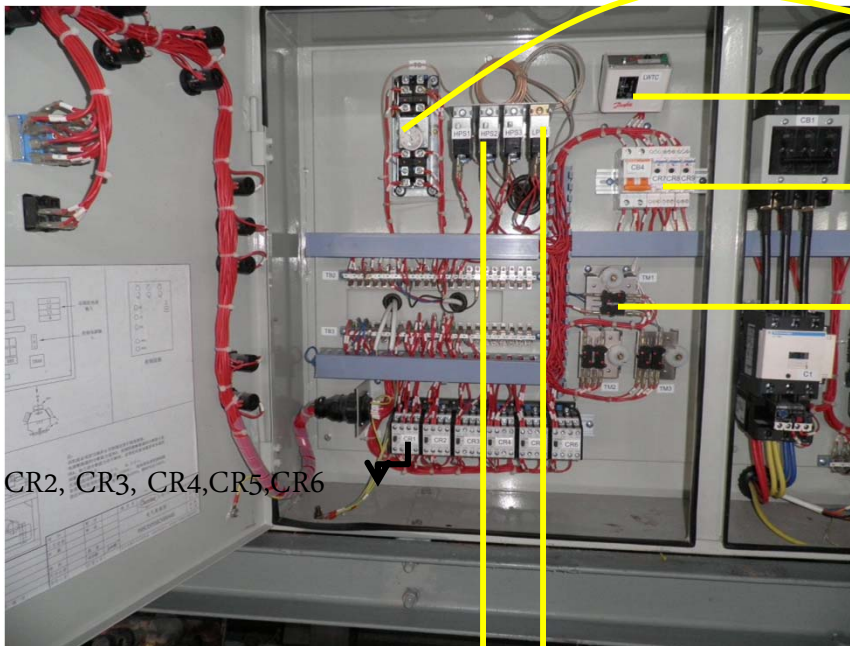
Low pressure line

Evaporator

Water Chilled outlet

Water Chilled inlet

Control Panel



TC : temperature Control

LWTC : Temperature switch

Contactor CR7, CR8, CR9

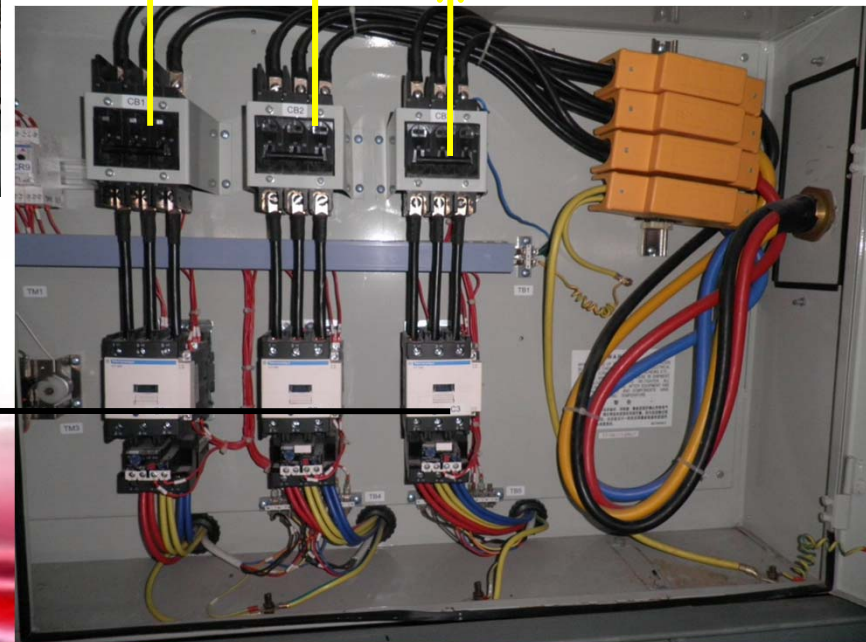
Timer TM1, TM2, TM3

CR1, CR2, CR3, CR4, CR5, CR6

HPS : HPS1, HPS2, HPS3

LPS

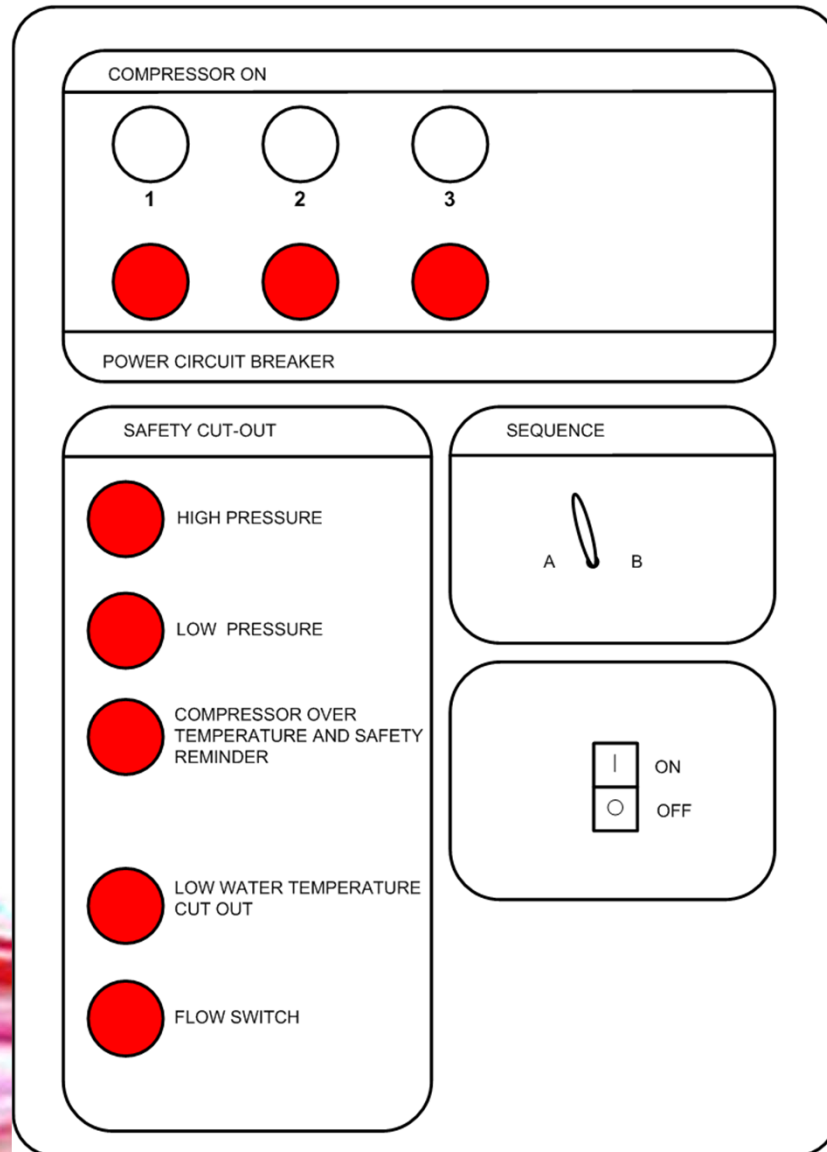
C1, C2, C3



CB1, CB2, CB3

Demo

Alarm Indicator



Keterangan

- Power circuit breaker alarm: CB1/CB2/CB3 Trip
- HP alarm: High pressure lebih dari 26 Bar
- LP alarm: Low pressure kurang dari 2 Bar
- Compressor over temperature and safety reminder alarm: kompresor kepanasan
- LWTC alarm: pembekuan pada keluaran water chilled SP: 0 °C
- Flow switch alarm: tidak aliran air pada inlet chilled

Control wiring

Perhatikan DRAWING WIRING CONTROL !!!

TERIMA KASIH